

Sampling & Dynamics

Notation: $f \in C_c^\infty(\mathbb{R}^d)$, $g \in C^\infty(\mathbb{R}^d)$

Recap $(P_t)_{t \geq 0}$ reversible Markov semigroup with generator $\mathcal{L}$ and stationary distribution π .

1. carré du champ operator Υ

$$\Upsilon(f, g) = \frac{1}{2} (\mathcal{L}(fg) - \mathcal{L}(f)g - f\mathcal{L}(g))$$

2. Iterated carré du champ operator Υ_2

$$\Upsilon_2(f, g) = \frac{1}{2} (\mathcal{L}\Upsilon(f, g) - \Upsilon(f, \mathcal{L}g) - \Upsilon(g, \mathcal{L}f))$$

- | | | |
|---|--|--|
| $\left\{ \begin{array}{l} 3. \end{array} \right.$ | <u>$CD(\alpha, \infty)$</u> | $\Upsilon_2(f, f) \geq \alpha \Upsilon(f, f)$ |
| $\left\{ \begin{array}{l} 4. \end{array} \right.$ | <u>LSI</u> | $\text{ent}_\pi(f^2) \leq 2 C_{LSI} E_\pi[\Upsilon(f, f)]$ |
| $\left\{ \begin{array}{l} 5. \end{array} \right.$ | <u>T_2</u> | $W_2^2(\nu, \pi) \leq 2 C_{T_2} K(\nu \ \pi) \quad \forall \nu \in \mathcal{P}_2(\mathbb{R}^d)$ |
| $\left\{ \begin{array}{l} 6. \end{array} \right.$ | <u>T_1</u> | $W_1^2(\nu, \pi) \leq 2 C_{T_1} K(\nu \ \pi) \quad \forall \nu \in \mathcal{P}_2(\mathbb{R}^d)$ |
| $\left\{ \begin{array}{l} 7. \end{array} \right.$ | <u>P_1</u> | $\text{var}_\pi f \leq C_{P_1} E_\pi[\Upsilon(f, f)]$ |

- Operations preserve functional inequalities
- Goal: Relations between functional inequalities

convex analysis $g: \mathbb{R}^d \rightarrow \mathbb{R}$ (twice continuously differentiable) ; g has a minimum at $x^* \in \mathbb{R}^d$

- $\left\{ \begin{array}{l} \bullet \text{ } \alpha\text{-strongly convex} : \Upsilon^2 g \geq \alpha \text{Id} \\ \bullet \text{ Gradient domination: } \underline{\|\nabla g(x)\|^2} \geq 2\alpha (g(x) - \inf g) \text{ (K inequality)} \\ \bullet \text{ Quadratic growth/stability: } g(x) - g(x^*) \geq \frac{\alpha}{2} \|x - x^*\|^2 \end{array} \right.$

Analogy: $(\mathcal{P}_2(\mathbb{R}^d), W_2)$, $\mu \mapsto K(\mu \| \pi)$ where $\pi \propto e^{-V}$ (V - some potential)
 $\mathcal{L}$ - Langevin generator $\Leftrightarrow (P_t)_{t \geq 0}$

- $\left\{ \begin{array}{l} \bullet K(\cdot \| \pi) \text{ } \alpha\text{-strongly convex} \Leftrightarrow \mathcal{L} \text{ satisfies } CD(\alpha, \infty) \text{ condition} \\ \bullet \text{ gradient domination for } K(\cdot \| \pi) \Leftrightarrow LSI \text{ for } \mathcal{L} \\ \bullet \text{ Quadratic growth for } K(\cdot \| \pi) \Leftrightarrow T_2 \end{array} \right.$

Techniques

- Taylor Expansion
- Lyapunov function
- duality

α strongly convex $\Rightarrow$ Gradient domination

$$\nabla^2 g \succeq \alpha \text{Id}$$

$$g(y) \geq g(x) + \langle \nabla g(x), y-x \rangle + \frac{\alpha}{2} \|y-x\|^2$$

choose $y = x^*$

$$g(x^*) - g(x) \geq \langle \nabla g(x), x^* - x \rangle + \frac{\alpha}{2} \|x^* - x\|^2 \geq -\|\nabla g(x)\| \|x^* - x\| + \frac{\alpha}{2} \|x^* - x\|^2$$

$$\geq -\frac{\|\nabla g(x)\|^2}{2\alpha}$$

$$\Rightarrow \|\nabla g(x)\|^2 \geq 2\alpha (g(x) - g(x^*)) \quad (\text{G.D} / \text{P.D})$$

Gradient domination $\Rightarrow$ Quadratic growth

$$\frac{d}{dt} \phi \circ \gamma_t < 0 \quad \text{for } \gamma_t \text{ gradient flow of } g$$

$$g(x) - g(x^*) \geq \frac{\alpha}{2} \|x - x^*\|^2$$

Idea:

$$\phi(x) = \sqrt{\frac{\alpha}{2}} \|x - x^*\| + \sqrt{g(x) - g(x^*)}$$

$$\dot{\gamma}_t = -\nabla g(\gamma_t)$$

$$|\langle a, b \rangle| \leq \|a\| \|b\|$$

$$\Rightarrow \frac{|\langle a, b \rangle|}{\|a\|} \leq \|b\|$$

$$\frac{d}{dt} \phi \circ \gamma_t = \frac{d}{dt} \left(\sqrt{\frac{\alpha}{2}} \|\gamma_t - x^*\| + \sqrt{g(\gamma_t) - g(x^*)} \right)$$

$$= \sqrt{\frac{\alpha}{2}} \frac{\langle \gamma_t - x^*, \dot{\gamma}_t \rangle}{\|\gamma_t - x^*\|} + \frac{\langle \nabla g(\gamma_t), \dot{\gamma}_t \rangle}{2 \sqrt{g(\gamma_t) - g(x^*)}}$$

$$= -\sqrt{\frac{\alpha}{2}} \frac{\langle \gamma_t - x^*, \nabla g(\gamma_t) \rangle}{\|\gamma_t - x^*\|} - \frac{\|\nabla g(\gamma_t)\|^2}{2 \sqrt{g(\gamma_t) - g(x^*)}}$$

$$\leq \sqrt{\frac{\alpha}{2}} \|\nabla g(\gamma_t)\| - \frac{\|\nabla g(\gamma_t)\|^2}{2 \sqrt{g(\gamma_t) - g(x^*)}}$$

$$= \|\nabla g(\gamma_t)\| \left(\sqrt{\frac{\alpha}{2}} - \frac{\|\nabla g(\gamma_t)\|}{2 \sqrt{g(\gamma_t) - g(x^*)}} \right)$$

$$\leq 0$$

$$\phi \circ x^* \leq \phi \circ x_0$$

$$\Rightarrow \sqrt{\frac{\alpha}{2}} \|x^* - x_0\| \leq \sqrt{g(x_0) - g(x^*)} \xrightarrow{\text{square}} \text{Quadratic Growth}$$

$$\nabla \rightarrow \nabla_{w_2} \quad \text{Otto-Villani Theorem}$$

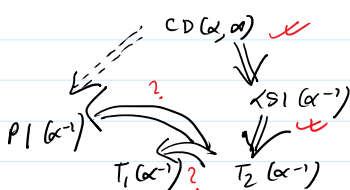
Caution: consider quadratic growth for g

$$g(x^* + \varepsilon y) - g(x^*) \geq \frac{\alpha}{2} \|\varepsilon y\|^2$$

$$\langle \varepsilon y, \nabla^2 g(x^*) \varepsilon y \rangle + o(\varepsilon^2) \geq \frac{\alpha}{2} \|\varepsilon y\|^2$$

$$\text{divide by } \varepsilon^2, \varepsilon \rightarrow 0 \quad \langle y, \nabla^2 g(x^*) y \rangle \geq \frac{\alpha}{2} \|y\|^2 \Rightarrow \nabla^2 g(x^*) \succeq \alpha \text{Id}$$

If g have $T_2 \not\Rightarrow g$ is α -strongly convex at the minimizer



$$T_2(\alpha^{-1}) \Rightarrow T_1(\alpha^{-1})$$

$$w_1 \leq w_2$$

Linearisation Principle

$$\kappa \propto ((1 + \varepsilon) \pi \| \pi)$$

$$\int (1 + \varepsilon) d\pi = 1$$

$$\begin{aligned}
&= \int (1+\epsilon f) \log(1+\epsilon f) d\pi \\
&= \int \left(\epsilon f + \frac{1}{2} \epsilon^2 f^2 + o(\epsilon^2) \right) d\pi(x) \\
&= \frac{1}{2} \epsilon^2 \int f^2 d\pi + o(\epsilon^2)
\end{aligned}$$

$$\Rightarrow \int f d\pi = 0$$

$w_2^2((1+\epsilon f)\pi, \pi)$ coupling b/w $(1+\epsilon f)\pi$ & π

$$\geq \int g(1+\epsilon f) d\pi + \int g^c d\pi \quad \forall g \in \mathcal{L}^1(\pi)$$

choose $g = \lambda \epsilon f$ (Reverse eng.)

$$\int g(1+\epsilon f) d\pi = \int (\lambda \epsilon f + \lambda \epsilon^2 f^2) d\pi = \lambda \epsilon^2 \int f^2 d\pi$$

$$\begin{aligned}
g^c(x) &= (\lambda \epsilon f)^c(x) = \inf_{y \in \mathbb{R}^d} \{ \|y - x\|^2 - \lambda \epsilon f(y) \} \\
&= \inf_{h \in \mathbb{R}^d} \{ \|h\|^2 - \lambda \epsilon f(x+h) \} \\
&= \inf_{h \in \mathbb{R}^d} \{ \|h\|^2 - \lambda \epsilon f(x) - \lambda \epsilon \langle \nabla f(x), h \rangle + o(\epsilon^2) \} \\
&\geq \frac{\lambda^2 \epsilon^2}{4} \|\nabla f(x)\|^2 - \lambda \epsilon f(x) - \frac{\lambda^2 \epsilon^2}{2} \|\nabla f(x)\|^2 + o(\epsilon^2) \\
&= \left(-\frac{\lambda^2 \epsilon^2}{4} \|\nabla f(x)\|^2 - \lambda \epsilon f(x) + o(\epsilon^2) \right)
\end{aligned}$$

$$\begin{aligned}
w_2^2((1+\epsilon f)\pi, \pi) &\geq \lambda \epsilon^2 \int f^2 d\pi - \frac{\lambda^2 \epsilon^2}{4} \int \|\nabla f(x)\|^2 d\pi - \lambda \epsilon \int f(x) d\pi + o(\epsilon^2) \\
&\geq \frac{(\int f^2 d\pi)^2}{\int \|\nabla f\|^2 d\pi} \epsilon^2 + o(\epsilon^2)
\end{aligned}$$

$$T_2(C) \quad 2C \mu((1+\epsilon f)\pi \| \pi) \geq w_2^2(\pi, (1+\epsilon f)\pi)$$

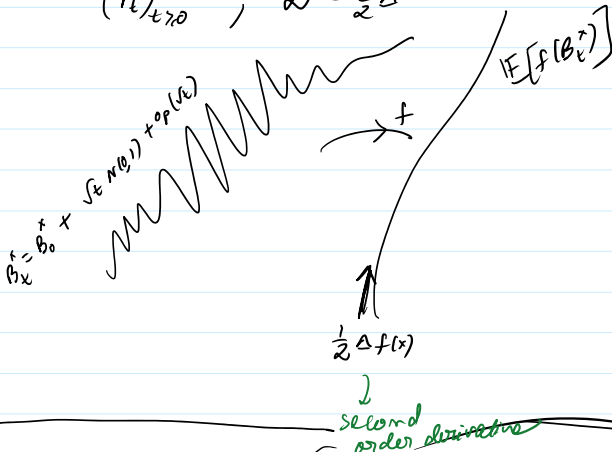
$$\Rightarrow 2C \left(\frac{\epsilon^2}{2} \int f^2 d\pi + o(\epsilon^2) \right) \geq \frac{(\int f^2 d\pi)^2}{\int \|\nabla f\|^2 d\pi} \epsilon^2 + o(\epsilon^2) \xrightarrow{\frac{1}{\epsilon^2}, \epsilon \rightarrow 0} C \int f^2 d\pi \geq \frac{(\int f^2 d\pi)^2}{\int \|\nabla f\|^2 d\pi}$$

$$f \neq 0 \Rightarrow (\int f^2 d\pi) \leq C \int \|\nabla f\|^2 d\pi \quad P1(C)$$

Proven various implications for Langevin diffusion
using duality, Taylor exp, asymptotic expansion

Goal: Algebraic properties of $\mathcal{L}$

eg: $dX_t = dB_t^x$
 $(P_t)_{t \geq 0}, \quad \mathcal{L} = \frac{1}{2} \Delta$



$$dX_t = \gamma'(t) dt \quad \gamma - \text{smooth curve in } \mathbb{R}^d$$

$$\gamma(0) = x$$

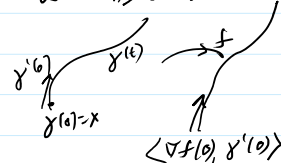
$$P_t f(x) = \mathbb{E}[f \circ \gamma(t)]$$

$$= f \circ \gamma(t)$$

$$\mathcal{L} f(x) = \frac{d}{dt} P_t f(x) = \langle \nabla f(x), \gamma'(0) \rangle$$

" $\mathcal{L}$ " is a directional derivative along $\gamma'(0)$

first order der.



Alg. Prop. • Leibnitz rule: $\rightarrow$ +linearity

$\downarrow$ second order derivative
 • Leibnitz rule $\times$
 • commutativity $\times$

$$\mathcal{L}(fg) = \mathcal{L}(f)g + f\mathcal{L}(g) + 2\Upsilon(f,g)$$

$$\nabla \mathcal{L} - \mathcal{L} \nabla = -(\nabla^2 v) \nabla f$$

Alg. Prop. $\langle \nabla f(0), \nabla g'(0) \rangle$
 • Leibnitz rule: $\nabla_v(fg) = (\nabla_v f)g + f(\nabla_v g)$ $\rightarrow$ +linearity charac. all ∇_v
 • commutativity: $\nabla_v \nabla_w f - \nabla_w \nabla_v f = 0$

GOAL: use these to prove the reln. b/w ∇ & $\mathcal{I}$.

Recap - Relation b/w Functional Inequalities

$$\begin{aligned}
 & - CD(\alpha, \infty) \\
 & \quad \Downarrow \\
 & \quad dSI(\alpha^{-1}) \\
 & \quad \Downarrow \\
 & \quad P_1(\alpha^{-1}) \\
 & T_1(\alpha^{-1}) \leq \frac{P_1(\alpha^{-1})}{T_2(\alpha^{-1})}
 \end{aligned}$$

For the Langevin diffusion
 $dx_t = -\nabla V(x_t)dt + \sqrt{2}dB_t$

- Q: How to extend this analysis to other diffusions?
- Algebraically: $P_t, \mathcal{L}, \Upsilon, T_2$

Product rule $\left[\begin{array}{l} \cdot \mathcal{L}(fg) \\ = \mathcal{L}(f)g + f\mathcal{L}(g) + 2\Upsilon(f,g) \end{array} \right.$

non commutativity $\left[\begin{array}{l} \cdot \nabla \mathcal{L} - \mathcal{L} \nabla = -\nabla^2 v \nabla \\ \text{(Langevin diffusion)} \end{array} \right.$

$\mathcal{L}$ D_v (directional derivative)

Goal: use these properties to generalise the relation b/w ∇ & $\mathcal{I}$

$\forall f, g$ in the domain of Υ

$$\int f(-\mathcal{L}g) d\pi = \int (-\mathcal{L}f)g d\pi = \int \Upsilon(f,g) d\pi =: \mathcal{E}(f,g) \quad \text{"E" Dirichlet Energy}$$

NOTE: We will deal only with "diffusion semigroups"

Defn: $(P_t)_{t \geq 0}$ diffusion semigroup if $\forall f, g \in \mathcal{L}^2(\pi)$ in the domain of Υ and all $\phi: \mathbb{R} \rightarrow \mathbb{R}$, the following holds:

$$\mathcal{L}(\phi \circ f) = \phi'(f)\mathcal{L}(f) + \phi''(f)\Upsilon(f,f) \quad \text{Itô correction}$$

chain rule $\rightarrow$ $\Upsilon(\phi \circ f, g) = \phi'(f)\Upsilon(f, g)$

In particular, discrete time / discrete state space Markov chain, we generally do not have a chain rule.

Warm up:

$$\begin{aligned}
 CD(\alpha, \infty) & \Rightarrow PI & CD(\alpha, \infty) \quad T_2 \geq \alpha T_1 & \Rightarrow \text{Var}_\pi f \leq \frac{1}{\alpha} \mathcal{E}(\Upsilon(f,f)) \\
 CD(\alpha, \infty) & \Rightarrow \alpha SI & & \text{or } \frac{1}{2} \mathcal{E}(f,f) \\
 & & & \text{ent}_\pi f^2 \leq \frac{1}{1-\alpha} \mathcal{E}(f,f)
 \end{aligned}$$

$$\text{ent}_\pi f^2 \leq \frac{1}{\alpha} \mathcal{E}(f, f)$$

Classical convex analysis

$g: \mathbb{R}^d \rightarrow \mathbb{R}$ C^2 , α strongly convex

$$\nabla^2 g \geq \alpha \text{Id}$$

consider $\dot{x}_t = -\nabla g(x_t)$ $x_0 \in \mathbb{R}^d$

$(x_0, g(x_0))$

 (Assume, $\inf g$ is attained at x^*)

define $E(t) := g(x_t) - g(x^*)$

Dissipation: $\frac{d}{dt} E(t) = \langle \nabla g(x_t), \dot{x}_t \rangle = -\|\nabla g(x_t)\|^2 =: -I(t)$

kinetic energy

kinetic energy dissipates exp-fast:

$$I(t) \leq e^{-2\alpha t} I(0)$$

Proof: $I'(t) = 2 \langle \nabla^2 g(x_t) \nabla g(x_t), \dot{x}_t \rangle$
 $= -2 \langle \nabla^2 g(x_t) \nabla g(x_t), \nabla g(x_t) \rangle$
 $\leq -2\alpha \langle \nabla g(x_t), \nabla g(x_t) \rangle$
 $= -2\alpha I(t)$

Gronwall's lemma $\Rightarrow I(t) \leq e^{-2\alpha t} I(0)$

gradient domination

$$E(0) - E(\infty) = \int_0^\infty E'(t) dt = \int_0^\infty I(t) dt \leq \int_0^\infty e^{-2\alpha t} I(0) dt = \frac{1}{2\alpha} I(0)$$

$$\Rightarrow E(0) \leq \frac{1}{2\alpha} \|\nabla E(0)\|^2$$

$$(\Leftrightarrow g(x_0) - g(x^*) \leq \frac{1}{2\alpha} \|\nabla g(x_0)\|^2)$$

CD(α , α) $\Rightarrow$ Poincaré

- $\dot{x}_t = ?$ (but we have a "flow" defined by P_t)

- define $E(t) := \text{var}_\pi(P_t f)$

(assume, $\int f d\pi = 0$)

$$= \int (P_t f)^2 d\pi$$

dissipation (clean out book keeping with ICT)

$$E'(t) = \int \partial_t (P_t f)^2 d\pi$$

$$= \int 2 P_t f \partial_t P_t f d\pi = \int 2 P_t f \mathcal{L} P_t f d\pi$$

$$= -2 \int (P_t f) (-\mathcal{L} P_t f) d\pi$$

$$= -2 \int \chi(P_t f, P_t f) d\pi$$

$$= -2 \mathcal{E}(P_t f, P_t f)$$

Dirichlet Energy

Dirichlet Energy decays exponentially fast

$$\mathcal{E}(P_t f, P_t f) \leq e^{-2\alpha t} \mathcal{E}(f, f)$$

$\downarrow$ show Poincaré

$$\chi(P_t f, P_t f) \leq e^{-2\alpha t} P_t \chi(f, f) \leftarrow \text{can be done even if } P_t \text{ does not have a stationary measure}$$

(Recall: $I'(t) \leq -2\alpha I(t) \Rightarrow I(t) \leq e^{-2\alpha t} I(0)$
 $\frac{d}{dt} \log I(t)$)

$$\begin{aligned} \partial_s \underbrace{p_s \Upsilon(p_{t-s} f, p_{t-s} f)} &= p_s \{ \mathcal{L} \Upsilon(p_{t-s} f, p_{t-s} f) - 2 \Upsilon(p_{t-s} f, \mathcal{L} p_{t-s} f) \} \\ &= p_s \mathcal{Q}_2(p_{t-s} f, p_{t-s} f) \\ &\geq_{C(\alpha, \alpha)} 2 \alpha \underbrace{p_s \chi(p_{t-s} f, p_{t-s} f)} \end{aligned}$$

$$\Rightarrow \boxed{\chi(P_t, R_t) \leq e^{-\alpha t} P_t \chi(f, f)} \Rightarrow E(P_t, R_t) \leq e^{-\alpha t} E(f, f)$$

(otherwise grow wall)

$$E(0) - E(\infty) = \int_0^{\infty} E'(t) dt = 2 \int_0^{\infty} E(\psi_t, \psi_t) dt \leq 2 \int_0^{\infty} e^{-2\alpha t} E(\psi_t) dt$$

$$\Rightarrow \text{var}_\pi f \leq \frac{1}{2} \varepsilon(f, f)$$

$$CD(\alpha, \infty) \Rightarrow \alpha S | (\alpha^{-1})$$

↕ whenever q obeys a chain rule (diffusion)

- ent_T f $\leq \frac{1}{2\alpha} \varepsilon(\log f, f)$ for $f > 0$, $f \in L^2(\pi)$
 f is in the dom of $\mathcal{L}$

define $E(t) := \text{ent}_\pi(P_t f) = \int P_t f \log(P_t f) d\pi$

$$\phi(x) = x \log x$$

$$= \int \phi(p_t f) d\pi$$

disposition $E'(t) = \int \phi(P_t f) \partial_t P_t d\pi$
 $= \int (1 + \log P_t f) \partial_t P_t d\pi$

$$\int \mathcal{L}_E f d\pi = \int \mathcal{L}_E f d\pi = \int (\mathcal{L}_E f) d\mathcal{L}^* \pi$$

$$\mathcal{L}^*_a = 0$$

$$SP_t + d\pi = \mathbb{E}_{x_t \sim \pi}[f(x_t)]$$

$$\int_t^T f_t f_t d\tau = 0 = E[f(x_0)]$$

$$= \int (\log p_{\epsilon t}) \mathcal{L} p_{\epsilon t} d\pi$$

$$= -\mathcal{E}(\log p_{\epsilon t}, p_{\epsilon t})$$

$\varepsilon(\log p_{t+1}, p_{t+1})$ decays exponentially fast

$$E(\log p_{t,f}, p_{t,f}) = \int \chi(\log p_{t,f}, p_{t,f}) d\pi \xrightarrow{\text{P.O.P.}} \int \chi(-2g) d\pi$$

$$\stackrel{\text{reversibility}}{=} \int \alpha(p_t \log p_t f, f) d\pi = \int \alpha(G, g)$$

$$\Rightarrow L \text{ is s.a.} \quad \Rightarrow L \text{ is s.a.} \quad \leq \int \sqrt{\alpha(L \log L + L \log L)} \sqrt{\alpha(L)} d\alpha$$

$$\gamma \leq \sqrt{\int f \gamma (p_t \log p_t + p_t \log p_t) d\pi}$$

$\alpha(\log f, f)$

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

$$\mathbb{E}(\log f, f)$$

$$\begin{aligned} \mathbb{E}(\log p_{t,f}, p_{t,f}) &\leq \exp(-\alpha t) \int \alpha(\log f, f) d\pi \int f \log p_{t,f}(\log p_{t,f}, \log p_{t,f}) d\pi \\ &\leq \exp(-\alpha t) \sqrt{\int \alpha(\log f, f) d\pi} \sqrt{\int \alpha(\log p_{t,f}, \log p_{t,f}) d\pi} \\ &\Rightarrow \sqrt{\mathbb{E}(\log p_{t,f}, p_{t,f})} \leq \exp(-\alpha t) \sqrt{\int \alpha(\log f, f) d\pi} \sqrt{\mathbb{E}(\log p_{t,f}, p_{t,f})} \end{aligned}$$

$$\Rightarrow \mathbb{E}(\log p_{t,f}, p_{t,f}) \leq \exp(-2\alpha t) \int \alpha(\log f, f) d\pi$$

$$\Rightarrow \mathbb{E}(\log p_{t,f}, p_{t,f}) \leq \exp(-2\alpha t) \mathbb{E}(\log f, f)$$

gradient dom.

$$E(0) - E(\infty) = \int_0^\infty E'(t) dt = \int_0^\infty \mathbb{E}(\log p_{t,f}, p_{t,f}) dt \leq \int_0^\infty e^{-2\alpha t} \mathbb{E}(\log f, f) dt$$

$$\Rightarrow \text{ent}_\pi f \leq \frac{1}{2\alpha} \mathbb{E}(\log f, f) = \frac{1}{2\alpha} \mathbb{E}(\log f, f)$$

$$C(\alpha, \infty) \Rightarrow \alpha SI(\alpha^{-1})$$

Brascamp-Lieb inequality what if $\nabla^2 v$ does not have a uniform lower bound?

$$\begin{aligned} \text{if } \pi \propto e^{-v}, \quad v \text{ is } C^2 \text{ strictly convex} \\ \mathcal{L} = \Delta - \langle \nabla v, \nabla \rangle \end{aligned}$$

$$\Rightarrow \text{Var}_\pi f \leq \mathbb{E}_\pi [\nabla f^T (\nabla^2 v)^{-1} \nabla f] \quad (\Rightarrow \text{if } \nabla^2 v \geq \alpha \text{Id} \rightarrow \frac{1}{\alpha} \mathbb{E}[\|\nabla f\|^2])$$

$$\bullet \dot{x}_t = -\nabla E(x_t)$$

↓ recond.

$$\dot{x}_t = -(\nabla^2 E(x_t))^{-1} \nabla E(x_t) = -\nabla_m E(x_t) \quad \langle, \rangle_m = \langle, \nabla^2 E \rangle$$